DECO7230 · DESIGN CONCEPT REPORT · SUBMISSION 1

XR Renovation Previewer

Redesigning the phone paint-visualiser as an embodied XR tool

Kaike Nehme · Semester 2, 2026 · Digital Prototyping and Extended Reality

01 THE APPLICATION, AND WHY IT NEEDS XR

CATEGORY — CREATION

The application being redesigned is the **phone renovation-preview app** — Dulux Visualizer, Houzz, IKEA Kreativ, Home Depot Project Color. These are interior-design and space-planning tools: the user creates a scheme for a room they are about to change.

They all fail in the same way. A candidate colour or finish is shown as a flat rectangle on a six-inch screen, rendered under generic lighting and detached from the room it is destined for. Yet the three variables that actually determine whether a renovation succeeds are all spatial:

Scale	Colour is perceived differently across a whole wall than in a thumbnail. A phone systematically misrepresents it.
Adjacency	What matters is how a candidate sits *next to* the floor, sofa or benchtop being kept — not in an isolated swatch grid.
Light	The variable that ruins real renovations. A phone shows one flat lighting condition; a room has many, and they change through the day.

Each is a property of *being in the space at full size*. This is therefore a genuine redesign rather than a port — XR supplies what the phone app structurally cannot.

02 USER TASKS AND GOALS

	TASK	GOAL
T1	Distinguish what is staying in the room from what is changing	Establish the constraint set that any scheme must satisfy — most renovations are not blank slates
T2	Evaluate a candidate finish against a fixed element, in place, at full size, under the room's own light	Judge a single choice with the context that actually decides it, rather than in isolation
T3	Assemble and compare complete schemes that all respect that same constraint	Commit to a whole-room direction instead of a sequence of unrelated individual colours

03 THE XR CONCEPT

Immersive environment. A domestic interior — one room. In the early prototypes this is a modelled virtual living room; in the final prototype it is the user's *own* room, seen through Meta Quest passthrough, so the fixed elements being designed around are real.

The central idea. You do not choose from a catalogue. You **pull a sample directly off something you are keeping** — the timber floor, the sofa — and that sample generates the range of finishes that work with it. The constraint that already exists in the room becomes the thing that produces the options, which removes both the menu and the choice paralysis in one move.

STEP 01	STEP 02	STEP 03	STEP 04	STEP 05
Mark	**Pull**	**Hold up**	**Tune**	**Commit**
Sweep an open hand across each surface to tag it as staying or changing.	Touch a kept surface and draw your hand away. A sample of it comes with you.	Raise it toward a changeable surface. As it nears, that surface previews the colour.	Twist your wrist to move through the range that works with the source. Warmer, cooler, deeper.	Release. Move away without releasing and the preview reverts.

There is no menu anywhere in that sequence. Lighting is controlled the same way — by touching a lamp or a wall switch, cycling warm bulb, cool bulb and daylight — so that every control is an object in the room rather than an interface laid over it.

04 XR INTERACTIONS AND AFFORDANCES

AFFORDANCE	WHAT THE USER DOES	WHY THE PHONE CANNOT DO THIS
Full-scale judgement	Stands in the room and sees a candidate across an entire wall	Colour perception is scale-dependent; a thumbnail is not a smaller version of the same experience
Physical adjacency	Holds a sample so it visually touches the floor being kept	A phone can only place them side by side in a grid, never in contact in real space
Real and variable light	Touches a lamp to switch between warm bulb, cool bulb and daylight	The app renders one fixed lighting condition, so the decision is made under light that will never exist
Proprioceptive comparison	Holds the sample at arm's length, moves it into shadow, into the corner	This is already how people judge samples in shops. A touchscreen cannot recruit the arm at all
Embodied parameter control	Rotates the wrist to travel a constrained colour range	Replaces a scrollable palette with a continuous physical movement through options that are already known to work
Viewpoint by locomotion	Walks to where the sofa sits and looks back at the wall	Colour and finish shift with viewing angle and distance; a fixed camera hides this entirely

05 IDEATION AND REFINEMENT

The initial idea was simply *"recolour the walls of a room in VR."* It reached its current form through two rounds of refinement.

Round 1 — low-fidelity physical prototype

A living room was built from paper, cardboard and playdoh in the Week 2 Studio — two hinged paper walls scribbled red and green, a bare kraft-card armchair, a playdoh sofa, a card table, two LED tea lights. Building it produced two unplanned findings:

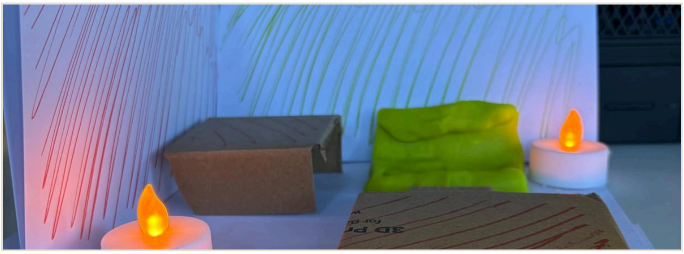

Plate 1. The same red paper reads warm orange beside the flame and cool pink away from it — one surface, two apparent colours.

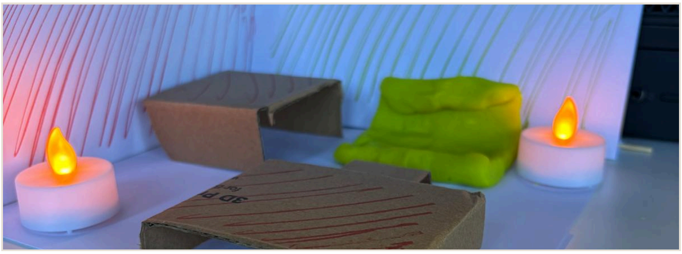

Plate 2. At eye level. The playdoh sofa was matched to the green wall and the card table to the red one, unprompted.

Light	The tea lights were an afterthought, and became the most useful element. They demonstrated the concept's central claim — that apparent colour depends on light — in cardboard, before any code existed. Light control was promoted from "out of scope" to a core interaction.
Matching	Furniture was coordinated to wall colours by hand while building, without being planned. The behaviour the concept is designed to support appeared spontaneously, which is weak but real support for the premise.

Round 2 — peer critique in Studio

Three requests came from peers, each phrased as a request for a button. Two had a better answer underneath:

Light on/off	**Adopted, reframed.** Implemented as a lamp you touch rather than a menu button, and extended to three states so that "does this green survive a warm bulb at 9pm?" becomes answerable.
Rotate furniture	**Adopted, deferred.** Initially read as an artefact of tabletop scale — you cannot walk around a model on a desk. Checking with participants established the intent was genuine layout change, so it becomes its own system in a later prototype rather than a control bolted onto this one.
Switch VR / MR	**Adopted, promoted.** Originally a one-way progression between prototypes; reframed as a live switch, letting a scheme be compared in the real room against an idealised virtual version of it.

The core loop was not challenged in either round and is carried forward unchanged.

06 INITIAL TESTING PLAN

What will be tested. The sample loop — pull from a kept surface, hold up to preview, twist to tune, release to commit — and the constraint mechanism that generates options from the kept surfaces. Scheme comparison, layout editing and passthrough are excluded from this first round.

	ASSUMPTION	WHAT WOULD INVALIDATE IT
A1	**Discoverability.** The loop can be performed without instruction	Participants tap or press surfaces, hunt for a menu, or need prompting before completing a single change
A2	**Constraint as help.** Generating options from what is staying reduces choice paralysis rather than causing frustration	Participants report the options as restrictive, ask for a full palette, or abandon the constraint when offered the chance

A2 is the decisive one. The entire concept rests on it, and it can genuinely fail — an invalidation would be a substantive finding rather than a setback.

Data to be collected

MEASURE	TYPE	METHOD
Time to first unprompted successful change	quantitative	Observer, stopwatch
Number of facilitator prompts required	quantitative	Observer tally
Whether twist-to-tune is discovered unprompted	boolean	Observation
Samples tried before committing to one	quantitative	Observer count
Confidence in the final choice, 1–5	quantitative	Post-task question
Whether the limited option set felt helpful or restrictive, and why	qualitative	Post-task question — **primary evidence for A2**
Reasons given for rejecting a candidate	qualitative	Think-aloud, recorded verbatim

Method. Think-aloud with no prior explanation of the interface, since discoverability cannot be measured once it has been described. Five minutes per participant, minimum five participants, run in the Studio session with peers and a tutor, screen-recorded from the headset with consent. Standardised instruments such as SUS and presence questionnaires are deliberately excluded at this stage — they are built for longer sessions and would report noise at five minutes.